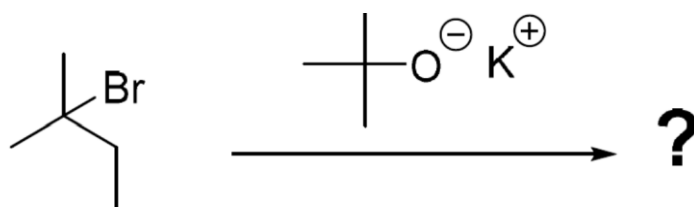
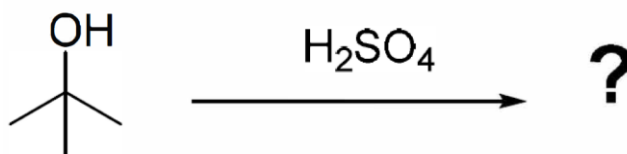


**QXU4004 – Engineering Chemistry**  
**Alkenes**

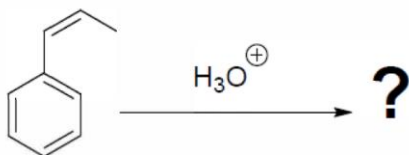
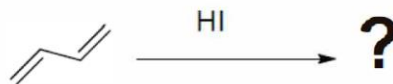
1. Why is E2 not easy for alcohols?
2. Draw the structure of 2-butene. HCl was induced to react with 2-butene. Write the mechanism of the reaction that proceeds of HCl with 2-butene. Give the name of the chlorinated product that is formed.
3. For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.



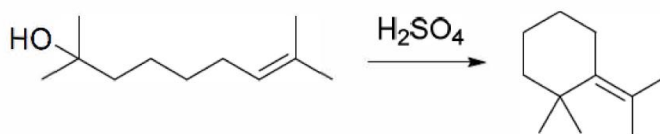
4. For the reaction below, draw the mechanism, give the structure of all the organic products that are formed indicating which one, if any, is formed as a major product stating why.



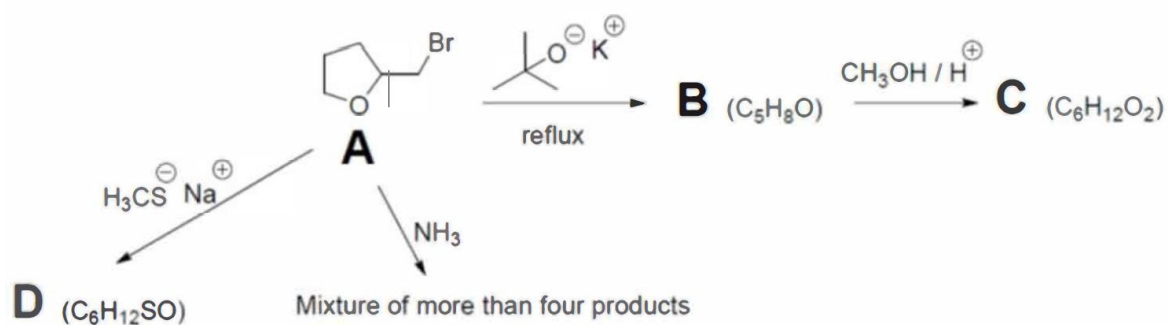
5. Propose all the possible products and mechanisms for the following reactions:



6. Draw the mechanism for the following reaction:



7. Consider the following reactions:



- Draw the structures of B, C and D
- Draw the mechanism of the reaction from A to B
- Draw the mechanism of the reaction from B to C, make sure you include relevant resonance forms as part of your mechanism
- Draw the mechanism of the reaction from A to D
- Explain using a mechanism, why the reaction between ammonia and A leads to a mixture of products